

3(a)(i)	F1 genotypes: AaBb x AaBb ; gametes: AB Ab aB ab (AB Ab aB ab) ; F2 genotypes: <table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab</td><td></td></tr><tr><td>AB</td><td>AABB awn</td><td>AABb awn</td><td>AaBB awn</td><td>AaBb awn</td><td></td></tr><tr><td>Ab</td><td>AABb awn</td><td>AAbb awn</td><td>AaBb awn</td><td>Aabb awn</td><td></td></tr><tr><td>aB</td><td>AaBB awn</td><td>AaBb awn</td><td>aaBB awn</td><td>aaBb awn</td><td></td></tr><tr><td>ab</td><td>AaBb awn</td><td>Aabb awn</td><td>aaBb awn</td><td>aabb no awn</td><td>;</td></tr></table> F2 phenotypes linked to genotypes (see Punnet Square) ; ratio of F2 phenotypes: 15:1 awn: no awn ;		AB	Ab	aB	ab		AB	AABB awn	AABb awn	AaBB awn	AaBb awn		Ab	AABb awn	AAbb awn	AaBb awn	Aabb awn		aB	AaBB awn	AaBb awn	aaBB awn	aaBb awn		ab	AaBb awn	Aabb awn	aaBb awn	aabb no awn	;	5
	AB	Ab	aB	ab																												
AB	AABB awn	AABb awn	AaBB awn	AaBb awn																												
Ab	AABb awn	AAbb awn	AaBb awn	Aabb awn																												
aB	AaBB awn	AaBb awn	aaBB awn	aaBb awn																												
ab	AaBb awn	Aabb awn	aaBb awn	aabb no awn	;																											
3(a)(ii)	autosomal dominant / epistasis ; I dihybrid	1																														

Question	Answer	Marks
3(b)	<i>any one from:</i> for seed dispersal ; idea of protection ; anchor into the ground ; obtain more, nutrients / water ;	1
3(c)(i)	<i>any four from:</i> 1 artificial selection / selective breeding / humans act as selection pressure ; 2 select (rice) plants / seeds / grains, with no awns ; 3 <i>idea of</i> breeding (selected) <u>plants</u> / self pollination of (selected) <u>plants</u> ; 4 seeds germinate / plants grow / offspring grow ; 5 repeat (selection and breeding) over <u>generations</u> ; 6 AVP ; e.g. <i>ref. to</i> genetic technology detail of pollination	4

Question	Answer	Marks
3(c)(ii)	any three from: 1 mutation / change in base sequence, (in <i>An-1</i>) or <i>An-1</i> <u>not</u> expressed ; 2 transcription factor has different, primary structure / tertiary structure / 3D shape or transcription factor <u>not</u>, produced / functional or transcription factor faulty ; 3 transcription factor does <u>not</u> bind to, the promoter / DNA ; 4 <u>decrease</u> in expression of gene(s) for awn development ; 5 <u>increase</u> in expression of gene(s) for number of grains produced ;	3

Question	Answer	Marks
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